

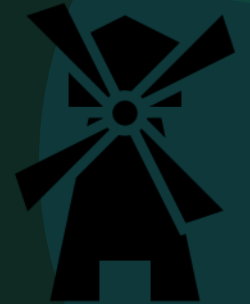
MSME CLEAN-ENERGY INNOVATION

AI-Enabled Hybrid Wind Turbine

Flexible Solar Film & Predictive Maintenance for MSMEs

Affordable, reliable, and sustainable power — engineered for the needs of small and medium enterprises.

[Team Name] • [Hackathon / Event Name] • [Date]



THE PROBLEM

Energy costs and downtime are eating into MSME margins



Rising energy costs

Grid tariffs and diesel-backup fuel costs consume a growing share of thin MSME operating margins.



Grid dependence & outages

Frequent, unpredictable power cuts halt production lines and damage sensitive equipment.



Unexpected downtime

Equipment fails without warning, forcing costly emergency repairs and missed delivery deadlines.



No unified monitoring

Energy use and machine health are tracked manually, if at all — problems are found too late.

OBJECTIVES

What we set out to build



Design a hybrid wind + flexible solar energy system sized for MSME power needs



Apply AI/ML to optimize energy generation and consumption in real time



Deploy IoT sensors for continuous, unattended equipment health monitoring



Enable predictive maintenance to cut unplanned downtime and repair costs



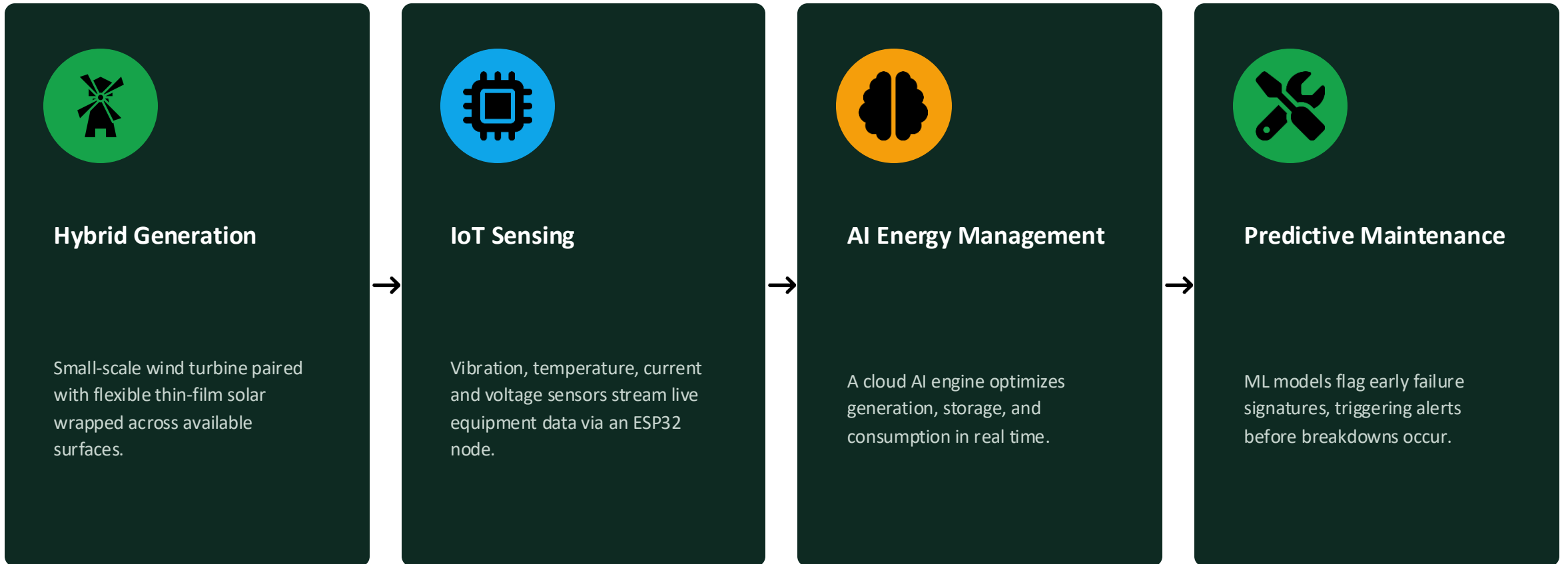
Deliver an affordable, scalable clean-energy solution with a fast payback period

Current approaches fall short

Approach	Limitation
✕ Standalone wind turbines	Intermittent output, large land footprint, no health monitoring
✕ Standalone rigid solar panels	Needs large flat roof/ground area; fragile; output drops on cloudy days
✕ Diesel generator backup	High fuel cost, emissions, noise — used only as a stopgap
✕ Manual maintenance schedules	Fixed-interval checks miss early failures and waste labor on healthy equipment

The common gap: none of these approaches combine dual-source generation with predictive, AI-driven equipment care.

One compact system, four integrated layers



What makes this different



Dual harvesting, one footprint

Wind and solar generation share a single compact installation instead of two separate systems.



Flexible, wrap-around solar film

Thin-film solar conforms to curved and vertical surfaces — no rigid panels, no large flat roof needed.



Edge + cloud AI

On-device TinyML handles instant anomaly checks; cloud analytics runs deeper predictive models.



Signature-based fault detection

Predictive maintenance reads vibration and thermal signatures, not just run-hours — catching issues generic timers miss.

Traditional vs. Our System

✗ Single energy source	✓ Hybrid wind + solar
✗ Manual inspection	✓ Continuous IoT monitoring
✗ Reactive repairs	✓ AI-predicted maintenance
✗ Rigid, bulky panels	✓ Flexible, low-profile film

From sensor data to actionable insight



Sensor readings are pre-filtered at the edge, synced to the cloud, scored by the AI model, and surfaced to operators as clear, actionable alerts.

Core hardware components



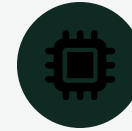
Small-scale hybrid turbine

Compact wind generator sized for rooftop / yard installation



Flexible thin-film solar

Lightweight, bendable PV film for curved and vertical surfaces



ESP32 microcontroller

Wi-Fi/Bluetooth-enabled edge node for sensor data collection



Vibration & temperature sensors

Continuous mechanical health monitoring on the turbine



Current & voltage sensors

Real-time tracking of energy generation and consumption



Li-ion battery + charge controller

Stores surplus energy and regulates safe charging

Software & tooling stack



Arduino IDE

Firmware development and flashing for the ESP32 edge node



Firebase / MQTT

Lightweight, low-latency pipeline for streaming sensor data to the cloud



Flutter

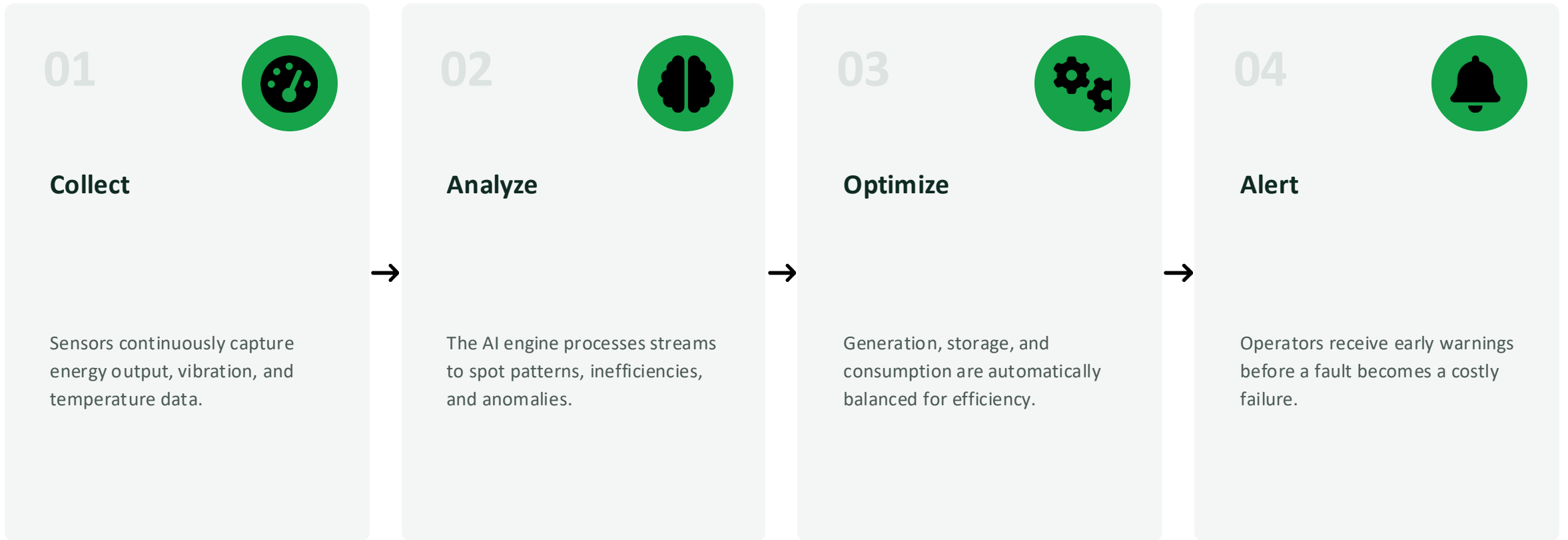
Cross-platform mobile dashboard for live monitoring and alerts



Python + TinyML

Model training in Python; compact TinyML models deployed to the edge

A continuous four-step loop



Catching failures before they happen

- 1 Vibration, temperature, and current sensors stream continuous readings from the turbine and solar array.
- 2 An ML model, trained on both normal and early-fault signatures, scores incoming data in real time.
- 3 Deviations that match known degradation patterns trigger a maintenance alert — well before a breakdown.
- 4 Every resolved alert feeds back into the model, sharpening future predictions.



Signals the model watches



Thermal drift

Rising bearing/motor temperature trend



Vibration spikes

Irregular frequency patterns vs. baseline



Output deviation

Power generation falling below expected curve

APPLICATIONS

Built for MSMEs across sectors



MSME Manufacturing

Small factories & production units



Farms & Agri-processing

On-site power for irrigation & processing



Workshops

Fabrication & repair workshops



Textile Units

Powering looms & finishing equipment



Rice & Agro Mills

Continuous milling power supply

Hardware sale + recurring subscription

Upfront



Hardware Sale

One-time purchase of the turbine, solar film, and IoT sensor kit — sized to the customer's power needs.

Recurring



AI Monitoring Subscription

Monthly or annual fee for the cloud dashboard, AI optimization, and predictive maintenance alerts.

Optional



Annual Maintenance Contract

Optional AMC covering scheduled servicing, part replacement, and priority support.

Target customers: individual MSMEs, industrial parks, and MSME clusters seeking shared clean-energy infrastructure.

Lower cost, higher reliability, cleaner operations



Lower energy costs

Reduced dependence on grid tariffs and diesel backup



Greater reliability

Continuous hybrid generation smooths supply gaps



Less downtime

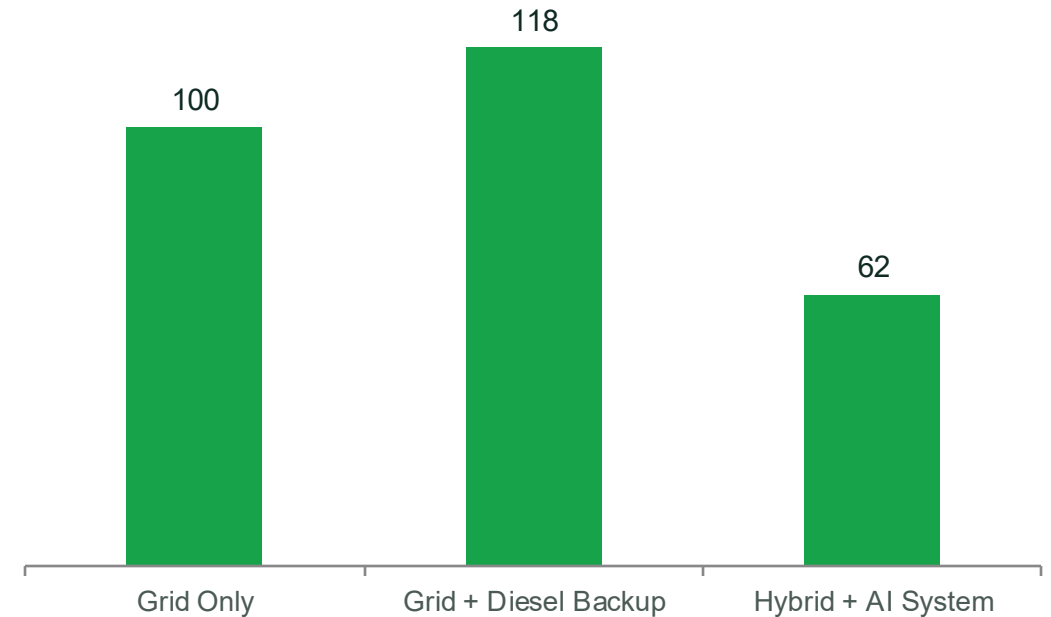
Early fault detection avoids emergency shutdowns



Sustainability

Cleaner power with a smaller carbon footprint

Illustrative annual energy-cost comparison



Figures are illustrative targets for the proposed system, not measured field data.

Where this goes next



Digital twin

A live virtual model of each installation for simulation and remote diagnostics.



Microgrid integration

Connecting multiple MSME sites for peer-to-peer energy sharing.



Active blade-pitch control

AI-driven blade adjustment to squeeze more output from variable wind.



Residential & rural expansion

Adapting the platform beyond MSMEs to homes and rural electrification.

CONCLUSION

A scalable clean-energy solution for MSMEs

By combining hybrid wind and flexible solar generation with AI-driven monitoring and predictive maintenance, this system gives MSMEs an affordable path to reliable, sustainable power — cutting energy costs and unplanned downtime at the same time.

- ✓ Dual-source generation in a compact, low-footprint system
- ✓ Real-time AI optimization of energy use
- ✓ Predictive maintenance that prevents costly failures
- ✓ A flexible business model built for MSME budgets



**Clean power.
Smart maintenance.
Built for MSMEs.**

Thank You

Questions & discussion welcome

[Team Name]

[Email / Phone]

[Institution / Organization]

